Deducing the inverse theorem for the multidimensional Gowers norms from the one-dimensional version

Terence Tao · 24 July 2015

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This week I have been at a Banff workshop "Combinatorics meets Ergodic theory", focused on the combinatorics surrounding Szemerédi's theorem and the Gowers uniformity norms on one hand, and the ergodic theory surrounding Furstenberg's multiple recurrence theorem and the Host-Kra structure theory on the other. This was quite a fruitful workshop, and directly inspired the various posts this week on this blog. Incidentally, BIRS being as efficient as it is, videos for this week's talks are already online.

As mentioned in the previous two posts, Ben Green, Tamar Ziegler, and myself proved the following inverse theorem for the Gowers norms:

> **Theorem 1 (Inverse theorem for Gowers norms)** Let $N \geq 1$ and $s \geq 1$ be integers, and let $\delta > 0$. Suppose that $f : \mathbf{Z} \to [-1,1]$ is a function supported on $[N] := \{1, \dots, N\}$ such that
>
> $$\frac{1}{N^{s+2}} \sum_{n,h_1,\dots,h_{s+1} \in \mathbf{Z}} \prod_{\omega \in \{0,1\}^{s+1}} f(n + \omega_1 h_1 + \dots + \omega_{s+1} h_{s+1}) \geq \delta.$$
>
> Then there exists a filtered nilmanifold G/Γ of degree $\leq s$ and complexity $O_{s,\delta}(1)$, a polynomial sequence $g : \mathbf{Z} \to G$, and a Lipschitz function $F : G/\Gamma \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta}(1)$ such that
>
> $$\frac{1}{N} \sum_n f(n) F(g(n)\Gamma) \gg_{s,\delta} 1.$$

There is a higher dimensional generalisation, which first appeared explicitly (in a more general form) in this preprint of Szegedy (which used a slightly different argument than the one of Ben, Tammy, and myself; see also this previous preprint of Szegedy with related results):

> **Theorem 2 (Inverse theorem for multidimensional Gowers norms)** Let $N \geq 1$ and $s, d \geq 1$ be integers, and let $\delta > 0$. Suppose that $f : \mathbf{Z}^d \to [-1,1]$ is a function supported on $[N]^d$ such that
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> $$\frac{1}{N^{d(s+2)}} \sum_{n,h_1,\dots,h_{s+1} \in \mathbf{Z}^d} \prod_{\omega \in \{0,1\}^{s+1}} f(n + \omega_1 h_1 + \dots + \omega_{s+1} h_{s+1}) \geq \delta. \quad (1)$$
>
> Then there exists a filtered nilmanifold G/Γ of degree $\leq s$ and complexity $O_{s,\delta,d}(1)$, a polynomial sequence $g : \mathbf{Z}^d \to G$, and a Lipschitz function $F : G/\Gamma \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta,d}(1)$ such that
>
> $$\frac{1}{N^d} \sum_{n \in \mathbf{Z}^d} f(n) F(g(n)\Gamma) \gg_{s,\delta,d} 1.$$

The $d = 2$ case of this theorem was recently used by Wenbo Sun. One can replace the polynomial sequence with a linear sequence if desired by using a lifting trick (essentially due to Furstenberg, but which appears explicitly in Appendix C of my paper with Ben and Tammy).

In this post I would like to record a very neat and simple observation of Ben Green and Nikos Frantzikinakis, that uses the tool of Freiman isomorphisms to derive Theorem 2 as a corollary of the one-dimensional theorem. Namely, consider the linear map $\phi : \mathbf{Z}^d \to \mathbf{Z}$ defined by

$$\phi(n_1, ..., n_d) := \sum_{i=1}^{d} (10N)^{i-1} n_i,$$

that is to say ϕ is the digit string base $10N$ that has digits $n_d ... n_1$. This map is a linear map from $[N]^d$ to a subset of $[d10^d N^d]$ of density $1/(d10^d)$. Furthermore it has the following "Freiman isomorphism" property: if $n, h_1, ..., h_{s+1}$ lie in $\mathbf{Z}$ with $n + \omega_1 h_1 + ... + \omega_{s+1} h_{s+1}$ in the image set $\phi([N]^d)$ of $[N]^d$ for all ω, then there exist (unique) lifts $\tilde{n} \in \mathbf{Z}^d, \tilde{h}_1, ..., \tilde{h}_{s+1} \in \mathbf{Z}$ such that

$$\tilde{n} + \omega_1 \tilde{h}_1 + ... + \omega_{s+1} \tilde{h}_{s+1} \in [N]^d$$

and

$$\phi(\tilde{n} + \omega_1 \tilde{h}_1 + ... + \omega_{s+1} \tilde{h}_{s+1}) = n + \omega_1 h_1 + ... + \omega_{s+1} h_{s+1}$$

for all ω. Indeed, the injectivity of ϕ on $[N]^d$ uniquely determines the sum $\tilde{n} + \omega_1 \tilde{h}_1 + ... + \omega_{s+1} \tilde{h}_{s+1}$ for each ω, and one can use base $10N$ arithmetic to verify that the alternating sum of these sums on any 2-facet of the cube $\{0,1\}^{s+1}$ vanishes, which gives the claim. (In the language of additive combinatorics, the point is that ϕ is a Freiman isomorphism of order (say) 8 on $[N]^d$.)

Now let $\tilde{f} : \mathbf{Z} \to [-1, 1]$ be the function defined by setting $\tilde{f}(\phi(n)) := f(n)$ whenever $n \in [N]^d$, with $\tilde{f}$ vanishing outside of $\phi([N]^d)$. If f obeys (1), then from the above Freiman isomorphism property we have

$$\frac{1}{N^{d(s+2)}} \sum_{n, h_1, ..., h_{s+1} \in \mathbf{Z}} \prod_{\omega \in \{0,1\}^{s+1}} \tilde{f}(n + \omega_1 h_1 + ... + \omega_{s+1} h_{s+1}) \geq \delta.$$

Applying the one-dimensional inverse theorem (Theorem 1), with δ reduced by a factor of $d10^d$ and N replaced by $d10^d N^d$, this implies the existence of a filtered nilmanifold G/Γ of degree $\leq s$ and complexity $O_{s,\delta,d}(1)$, a polynomial sequence $g : \mathbf{Z} \to G$, and a Lipschitz function $F : G/\Gamma \to \mathbf{R}$ of Lipschitz constant $O_{s,\delta,d}(1)$ such that

$$\frac{1}{N^{d(s+2)}} \sum_{n \in \mathbf{Z}} \tilde{f}(n) F(g(n)\Gamma) \gg_{s,\delta,d} 1$$

which by the Freiman isomorphism property again implies that

$$\frac{1}{N^{d(s+2)}} \sum_{n \in \mathbf{Z}^d} f(n) F(g(\phi(n))\Gamma) \gg_{s,\delta,d} 1.$$

But the map $n \mapsto g(\phi(n))$ is clearly a polynomial map from $\mathbf{Z}^d$ to G (the composition of two polynomial maps is polynomial, see e.g. Appendix B of my paper with Ben and Tammy), and the claim follows.

> **Remark 3** This trick appears to be largely restricted to the case of boundedly generated groups such as $\mathbf{Z}^d$; I do not see any easy way to deduce an inverse theorem for, say, $\bigcup_{n=1}^{\infty} \mathbb{F}_p^n$ from the $\mathbf{Z}$-inverse theorem by this method.

> **Remark 4** By combining this argument with the one in the previous post, one can obtain a weak ergodic inverse theorem for $\mathbf{Z}^d$-actions. Interestingly, the Freiman isomorphism argument appears to be difficult to implement directly in the ergodic category; in particular, there does not appear to be an obvious direct way to derive the Host-Kra inverse theorem for $\mathbf{Z}^d$ actions (a result first obtained in the PhD thesis of Griesmer) from the counterpart for $\mathbf{Z}$ actions.

Comments (1)



Anonymous · 3 August 2015

Great!